Catholic Junior College

JC2 Preliminary Examinations

Higher 1

CANDIDATE NAME	
CLASS	**2T**

CHEMISTRY

8873/02

Paper 2 Structured Questions

27 August 2021

2 hours

Candidates answer on the Question Paper.
Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** the questions.

Section B
Answer **one** question.

The use of an approved scientific calculator is expected where appropriate.
You are reminded of the need for good English and clear presentation in your answers.

A Data Booklet is provided.

The number of marks is given in brackets [ ] at the end of each question or part of the question.

Section A	Q1	20
	Q2	6
	Q3	7
	Q4	11
	Q5	10
	Q6	6
Section B	Q7	20
	Q8	20
Paper 2		80
Paper 1		30
Percentage		
Grade		

1(a) One of the industrial stages of producing fertilisers involves ammonia and oxygen reacting over a platinum catalyst at 800 ^oC. The products are nitric oxide (NO) and steam.

(i) Write a balanced equation for this reaction.

.... [1]

(ii) State the type of catalysis involved. Briefly explain how it works.

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.... [3]

(iii) The nitric oxide formed from the reaction is a pollutant gas. This gas is also present in the exhaust of vehicles as it is formed via this reaction:

$$N_2(g) + O_2(g) \rightarrow 2NO(g) \qquad \Delta H = +180 \text{ kJ mol}^{-1}$$

Suggest two reasons why this reaction can occur in car engines.

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.... [2]

(iv) Suggest why the reaction in **(iii)** is endothermic.

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.... [1]

(v) State the two other pollutants commonly present in exhaust gas.

.... [1]

(b) Nitric oxide can be oxidised to form NO_2 according to the following equation:

$$NO(g) + \frac{1}{2}O_2(g) \rightarrow NO_2(g) \qquad \Delta H = -57.0 \text{ kJ mol}^{-1}$$

(i) Draw the dot-and-cross diagram of NO_2.

[1]

(ii) Define standard enthalpy of formation of NO_2.

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....

.... [1]

(iii) Using information in **(a)(iii)** and **(b)**, calculate the standard enthalpy change of formation of $NO_2(g)$.

[2]

(c)(i) Nitrogen exhibits a range of oxidation numbers in its compounds.
Table 1.1 shows the oxidation state of nitrogen in different species. Fill in the blanks with the appropriate oxidation states.

Species	Oxidation state of nitrogen
NO_3^-	+5
NO_2	+4
NH_4^+	–3
NH_2OH	
N_2O	
NO	
N_2	0

Table 1.1

[1]

Nitric acid, HNO_3, can be reduced to different nitrogen-containing products, depending on the choice of reducing agent.
For example, copper metal can reduce HNO_3 to NO_2. However, aluminium metal reduces HNO_3 to NH_4^+.

(ii) With reference to the *Data Booklet*, write a balanced equation for the reaction between aluminium and nitric acid.

.... [2]

(iii) With reference to the change in oxidation numbers of nitrogen when nitric acid reacts with Cu metal and when nitric acid reacts with A*l* metal, identify the metal that is the weaker reducing agent.

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.... [2]

Fe^{3+}(aq) is an oxidising agent and its relevant reduction half equation is shown below:

$$Fe^{3+}(aq) + e^{-} \rightarrow Fe^{2+}(aq)$$

Fe^{3+} can oxidise NH_2OH to form one of the nitrogen-containing products listed in **Table 1.1**. The identity of the product can be found by determining the number of electrons lost by nitrogen in NH_2OH and hence finding the oxidation state of nitrogen in the product formed.

In an experiment, 25.0 cm^3 of 0.100 mol dm^{-3} NH_2OH required 21.70 cm^3 of 0.230 mol dm^{-3} of Fe^{3+}(aq) for complete reaction.

(iv) Calculate the amount (in moles) of Fe^{3+} that reacts with one mole of NH_2OH and hence determine the number of moles of electrons lost by one mole of NH_2OH in the reaction.

[2]

(v) With reference to **Table 1.1**, identify the product of the reaction.

.... [1]

[Total: 20]

2 Nitrosyl chloride, NOC*l*, is a pungent orange gas used commercially as a bleaching agent and in synthetic detergents. Upon heating, it decomposes according to the equation below.

$$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g) \qquad \Delta H < 0$$

A sample of 2.0 mol of pure NOC*l*, was heated at ***T*** °C in a sealed 2.0 dm^3 vessel. When equilibrium was reached, 25% of the NOC*l* gas had decomposed

(i) Write an expression for the equilibrium constant, K_c for this reaction.

[1]

(ii) Calculate the equilibrium constant, K_c at ***T*** °C and state its units.

[2]

(iii) State and explain how the extent of decomposition of NOC*l* gas would be affected by an increase in temperature.

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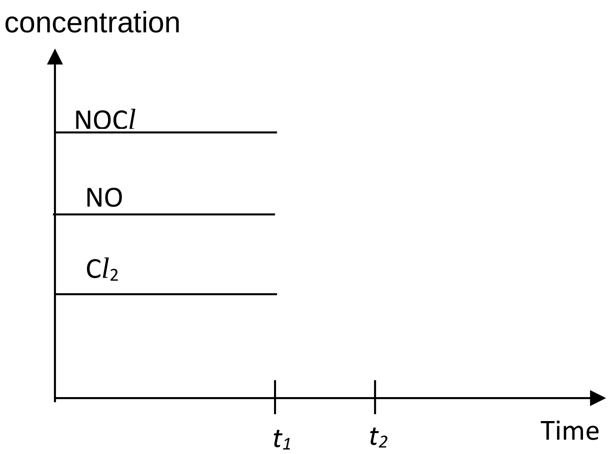
.... [2]

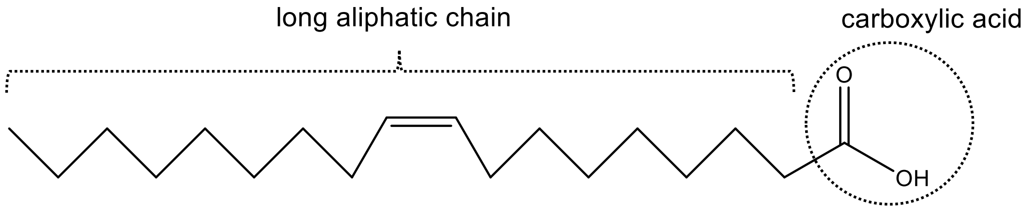
(iv) At time t_1, more NOC*l* gas was pumped into the vessel while keeping the temperature constant.
Complete the diagram below, indicating the effect of this change on the concentrations of NOC*l*, NO and Cl_2 when the new equilibrium is established at t_2.



[1]

[Total: 6]

3(a) Fatty acids are carboxylic acids with long aliphatic chains, which are either saturated or unsaturated. An example of a fatty acid is *oleic acid,* $CH_3(CH_2)_7CH{=}CH(CH_2)_7CO_2H$, shown below:



oleic acid

"Essential fatty acids" are required for good health. However, these acids cannot be produced in sufficient quantity in the human body and hence must be obtained from food. Such fatty acids are widely distributed in plant oils.

To measure the free fatty acid content present in plant oils, solutions of fatty acids in ethanol are usually titrated with potassium hydroxide, KOH solution.
Acid value (AV) is the total mass of KOH (in mg) required to neutralise the free acids present naturally in 1 g of oil.

AV is calculated using the following formula:

$$\text{Acid value (AV)} = \frac{\text{mass of KOH used (in mg)}}{\text{mass of sample (in g)}}$$

The table below shows the acid value of some common oils.

Oil	**Acid value (AV) (mg KOH / g sample)**
Canola	0.071
Soya	0.60
Virgin olive oil	6.6
Used frying oil	31

Table 3.1

A plant oil with low AV is a good dietary oil. When oil is subjected to prolonged high temperatures, triglycerides (a type of fat) in oil are converted into fatty acids, leading to an increase in the AV value.

10 g of an olive oil sample is dispersed in ethanol and made up to a final volume of 60.90 cm^3. The mixture is titrated with 50 cm^3 of 0.1 mol dm^{-3} KOH.

(i) Calculate the acid value (AV) of this sample of olive oil. (M_r of KOH = 56.1).

[1]

(ii) By comparing your answer in **(i)** with relevant values in **Table 3.1,** comment on the quality of the olive oil sample and suggest a possible reason for the difference.

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.... [1]

(iii) Assume that the acid is the only free fatty acid present in the sample of olive oil. Given that oleic acid and KOH react in a 1:1 mole ratio, calculate the initial concentration of oleic acid present in the final volume of olive oil sample.

[1]

(b)(i) Oils do not have pH values because the fatty acids remain undissociated in the oil. Only aqueous solutions of acids have pH values. Define pH.

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.... [1]

(ii) An aqueous sample of monobasic acid **X,** which has an identical concentration as the oleic acid calculated in **(a)(iii),** has a pH of 4.97. Calculate the concentration of H^+ in the oil sample.

[1]

(iii) Using the concentration values calculated in **(a)(iii)** and **(b)(ii)**, explain whether the aqueous acid **X** is a strong or weak acid.

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.... [1]

(iv) From the list of indicators given below, identify the **most suitable** indicator for the titration between acid **X** and KOH(aq).

	indicators	**pH range**
1	Bromocesol green	3.8 – 5.4
2	Bromothymol blue	6.5 – 7.0
3	Phenol red	6.8 – 8.4
4	Phenolpthalein	8.2 – 10.0

Suitable indicator: .... [1]

[Total: 7]

4(a) The following table contains the values of lattice energies of MgO and AgI:

Compound	Experimental lattice energy / kJ mol^{-1}	Theoretical lattice energy / kJ mol^{-1}
MgO	-3791	-3795
AgI	-889	-808

(i) Explain why the lattice energy of MgO is considerably larger in magnitude than AgI.

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.... [2]

(ii) There is close agreement between the experimental and theoretical values of lattice energy for MgO but not for AgI. Suggest a reason for this.

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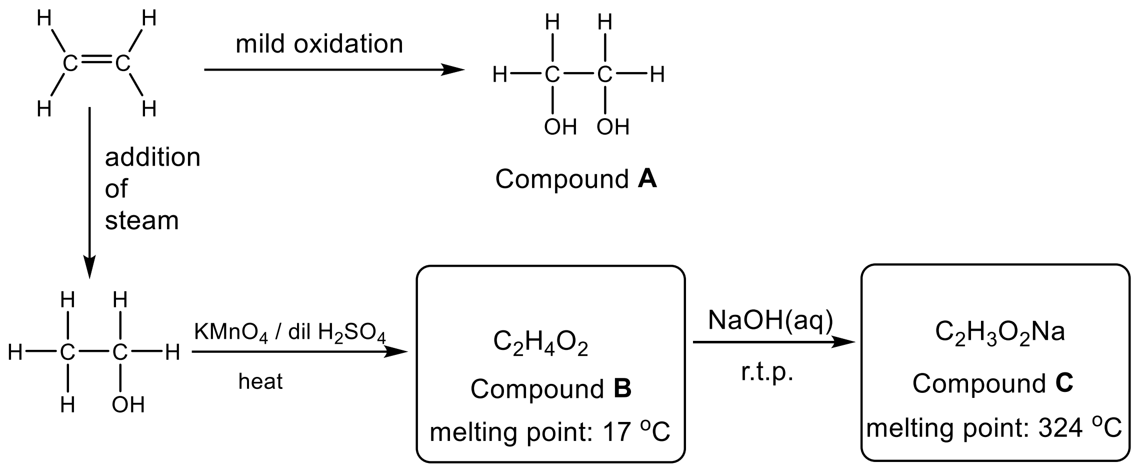
.... [1]

(b)



(i) Give the IUPAC name for **compound A**.

.... [1]

(ii) **Compound A** reacts with terephthalic acid to form a polyester called poly(ethylene terephthalate), also known as PET. Draw the structure of terephthalic acid and give its IUPAC name.

[2]

(iii) State the conditions necessary for such an esterification reaction to occur.

.... [1]

(iv) Draw the displayed formula for **compound B**, $C_2H_4O_2$, and clearly indicate the shape around each carbon atom.

[2]

(v) In terms of structure and bonding, explain the difference in melting point between compounds **B** and **C**.

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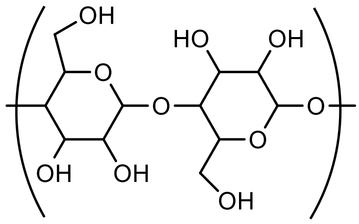
.... [2]

[Total: 11]

5(a) Frequent testing for Covid-19 requires swabs to be readily available for specimen collection. The traditional cotton tipped-swab is cost effective and safe. The repeating unit of cotton is shown below:



However, cotton swabs have recently been replaced by 'flocked nylon swabs' for better specimen collection. The tip of a 'flocked' swab arranges the nylon fibres in short parallel strands that make the swab tip more like a 'brush' that is capable of collecting more sample for testing. It is also better at releasing the sample during testing compared to the cotton swab. Samples collected from the nose or throat are generally water-based.

(i) One of the monomers that forms nylon-6,6 is shown below. In the space beside it, draw the skeletal structure of the other monomer of nylon-6,6.

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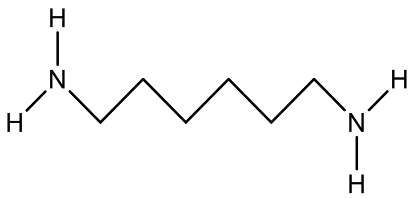

[1]

(ii) State the type of polymerisation involved in nylon-6,6.

.... [1]

(iii) Nylon-6,6 is an example of a *copolymer*. What does the term *copolymer* mean?

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.... [1]

(iv) With reference to the type of bonding involved, suggest a reason why cotton swabs are less preferred for use in sample testing compared to the nylon swab.

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.... [2]

(b)(i) Draw a section of poly(propene), showing three repeat units.

[1]

(ii) To what homologous series does poly(propene) belong?

.... [1]

(iii) When a deep wound is repaired by surgery and needs a prolonged period for healing, a mesh made of poly(propene) is inserted below the muscle tissue. The wound is less likely to reopen and the repair is stronger as compared to using a mesh made of polyester. Suggest why a poly(propene) mesh is stronger than a polyester mesh for use in this situation.

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.... [1]

(iv) Members of the homologous series you have given in **(ii)** are considered to be able to undergo two different kinds of reactions. Explain why neither of them can take place on a poly(propene) mesh inserted in living body tissues.

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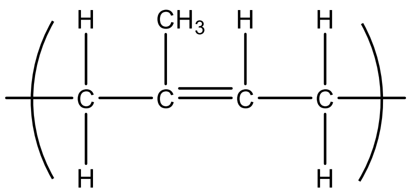
.... [2]

[Total: 10]

6 Latex gloves are very common in hospitals where they protect from infection while still allowing sensitive touch.

The repeat unit for latex (natural rubber) is shown below. About 10 000 units join together to form natural rubber.



(i) Draw the repeat unit in skeletal form, showing the cis-configuration.

[1]

(ii) The monomer unit for natural rubber is 2-methylbut-1,3-diene. Draw the structure of the monomer.

[1]

(iii) Liquid latex, can be solidified and strengthened by a process called *vulcanisation*. The term *vulcanised* comes from the name Vulcan, the god of volcanoes because of the sulphur, heat and pressure (similar to what are found in volcanoes) used in the solidification of latex. Latex is a thermoplastic polymer but vulcanised rubber is a thermoset polymer. State the bonding present in vulcanised rubber that is absent in latex.

.... [1]

(iv) Vulcanised rubber is used to make the tyres of vehicles because the material is durable and resistant to heat. In terms of structure and bonding, explain why vulcanised rubber is resistant to heat.

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.... [2]

(v) The chemical process of vulcanisation breaks the alkene groups in latex to allow for addition reaction between chains. Sulfur atoms hold adjacent chains together.

Draw a possible structure of vulcanised rubber, showing clearly how the S atoms hold adjacent chains together.

[1]

[Total: 6]

Section B

Answer **one** question in the spaces provided.

7(a) Sulfur belongs to Group 16 and Period 3 in the Periodic Table.

(i) Describe the trend in the first ionisation energies of the elements down Group 16. Explain your answer.

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.... [2]

Like sulfur, selenium also belongs to Group 16 but is in Period 4 in the Periodic Table. Selenium trioxide, SeO_3, is a white solid that readily absorbs moisture in the air.

(ii) Is SeO_2 expected to be an acidic or basic oxide?

.... [1]

(iii) Write an equation to show the reaction when SeO_3 reacts with moisture in the air.

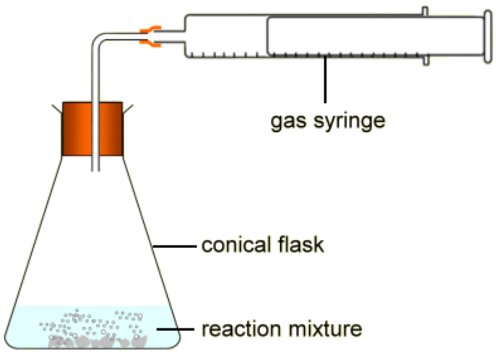
.... [1]

(b) **(i)** Draw a labelled diagram showing the orbital overlap between the carbon atom and an oxygen atom in carbon dioxide, O=C=O. (Just show the overlap between C=O.)

[2]

Carbon dioxide is released when an excess of powdered sodium carbonate was allowed to react with 100 cm^3 of 0.100 mol dm^{-3} of $HCl(aq)$.

The experimental set up is shown here:



The balanced equation for the reaction is:

$$2HCl(aq) + Na_2CO_3(s) \rightarrow 2NaCl(aq) + CO_2(g) + H_2O(l)$$

The volume of carbon dioxide gas liberated was measured at regular intervals and the results are tabulated in **Table 7.1** below.

Time / s	5	10	20	30	40	50	60	70
Volume of CO_2 / cm^3	28	50	77	96	107	115	**V**	**V**

Table 7.1

The reaction stopped at 60 seconds and the maximum volume, **V**, of $CO_2(g)$ was collected at that time.

(ii) Calculate the value of **V**, which is the volume of $CO_2(g)$ liberated at completion of the reaction. (Assume no loss of gas during the experiment and that the reaction took place at room temperature and pressure.)

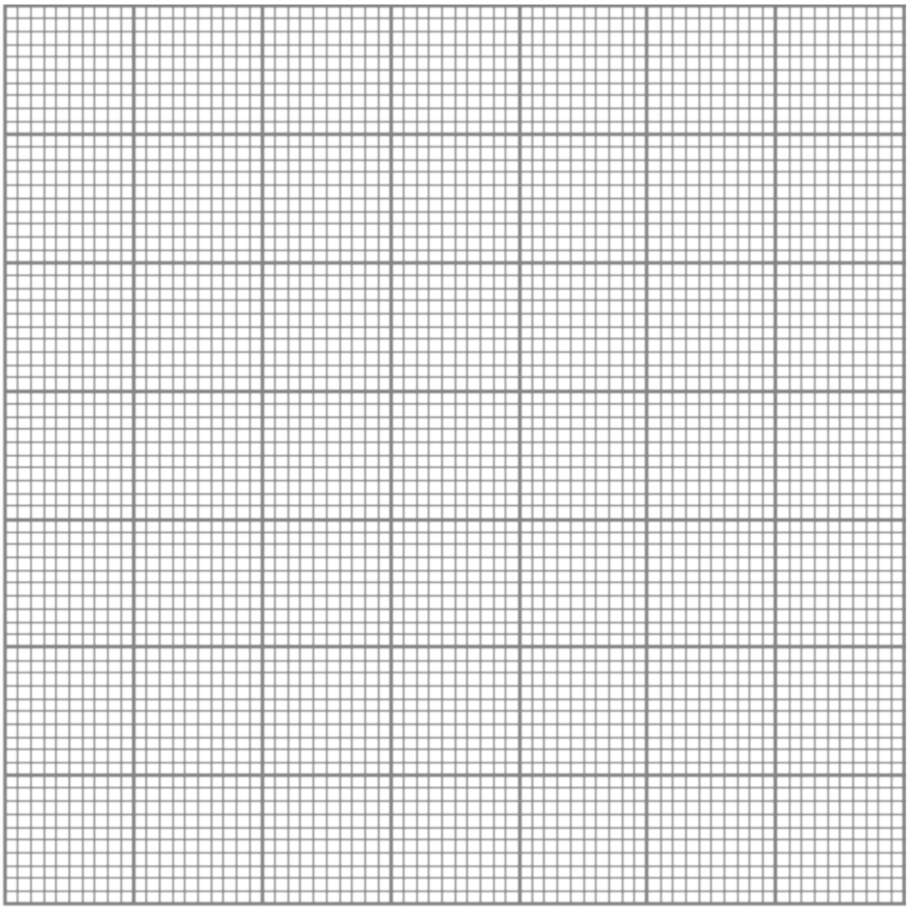
[2]

Table 7.1 is reproduced here for your convenience.

Time / s	5	10	20	30	40	50	60	70
Volume of CO_2 / cm^3	28	50	77	96	107	115	**V**	**V**

Table 7.1

(iii) Plot a graph of volume of CO_2 against time (up to 70 s) for the experiment.



[2]

(iv) The product-time graph above gives an indication of the order of reaction with respect to HCl(aq). Use your graph to determine the order of reaction with respect to HCl(aq), showing your construction lines clearly.

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.... [2]

(v) Considering the rate equation of the reactions,

$$rate = k[HCl]^n,$$

Explain why an increase in temperature will increase the rate constant, *k*.

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.... [3]

(vi) The experiment occurs fairly quickly. If the experiment was repeated at the same temperature, suggest a simple modification that will slow the reaction down without affecting the total volume of CO_2 collected.

.... [1]

(c) **(i)** State Le Chatelier's Principle.

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.... [1]

(ii) Yellow chromate (VI) ions, CrO_4^{2-} and orange dichromate (VI) ions, $Cr_2O_7^{2-}$ exist in equilibrium in aqueous solution as shown by the expression below.

$$2CrO_4^{2-}(aq) + 2H^+(aq) \rightleftharpoons Cr_2O_7^{2-}(aq) + H_2O(l)$$

Use Le Chatelier's Principle to describe and explain what would be observed when some NaOH(aq) was added to the aqueous solution.

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.... [3]

[Total: 20]

8(a) (i) The first ionisation energy generally increases across Period 3. Yet the first ionisation energy of sulfur is lower than the first ionisation of phosphorus which is the element preceding sulfur.
Explain why sulfur has a lower first ionisation energy than phosphorus.

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.... [2]

(ii) Aluminium and phosphorus react with chlorine to form $AlCl_3$ and PCl_5 respectively. Write equations to show what happens when samples of each of these chlorides are added separately to water. Briefly describe the reactions.

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.... [3]

8(b) Use of *Data Booklet* is relevant to this question.

A small cylinder containing propyne ($CH_3C{\equiv}CH$) fuel was used to heat a container of water. The cylinder was weighed before and after the experiment to determine the mass of propyne used. This process was known to be only 80% efficient. The following data was recorded.

Mass of cylinder before heating = 504.66 g
Mass of cylinder after heating = 502.16 g
Mass of water heated = 750 g
Initial temperature = 27.0 °C
Final temperature = 53.8 °C

(i) Calculate the heat energy absorbed by the water and hence calculate the enthalpy change of combustion of propyne.

[4]

(ii) Construct the reaction pathway diagram for the combustion of propyne. Label the E_a and ΔH clearly.

[2]

(c) **(i)** Define the term *bond energy***.**

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.... [1]

(ii) When propyne was reacted with H_2 (hydrogenation reaction) over a suitable catalyst, propane was formed.

$$H{-}C{\equiv}C{-}CH_3\,(g) + 2H_2\,(g) \longrightarrow CH_3CH_2CH_3\,(g)$$

Use relevant bond energy values from the *Data Booklet* to calculate the enthalpy change of the above hydrogenation reaction**.**

[2]

(iii) The enthalpy change of combustion of hydrogen is given as –286 kJ mol^{-1}.
Let the enthalpy change of combustion of propyne calculated in **(b)(i)** be **B**.
Let the enthalpy change of reaction of hydrogenation of propyne calculated in **(c)(ii)** be **C**.

In terms of **B** and **C**, write a mathematical equation that shows how the the enthalpy change of combustion of propane can be calculated from the above information.

$$CH_3CH_2CH_3(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$$

ΔH_c of propane = ... [1]

(d) The phosphate buffer system in the human body consists of $H_2PO_4^-$ and HPO_4^{2-} ions. The phosphate buffer system has a pH of 6.8 at its maximum buffering capacity, which is not far from the normal pH of 7.4 in the body fluids, making it an effective buffer for intracellular fluids.

(i) Define *buffer* solution.

....

.... [1]

(ii) Identify the acidic and basic components in the phosphate buffer system.

Acid: ....

Base: .... [1]

(iii) At maximum buffer capacity, the pH of the buffer system is equal to the pK_a value of $H_2PO_4^-$. Calculate the acid dissociation constant, K_a, of $H_2PO_4^-$.

[1]

(iv) With the use of relevant equations, show how the $H_2PO_4^-$ / HPO_4^{2-} buffer system works in the human body.

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.... [2]

[Total: 20]